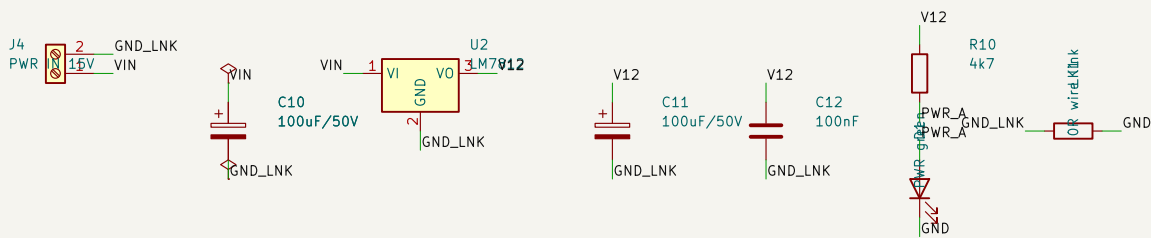


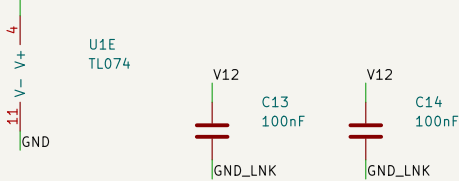
Bose Companion 5 sub crossover -- mono--summing 2nd--order Sallen--Key low--pass.

Single +12 V rail, no negative supply. All AC referenced to VGND = 6 V (A3 buffers the R8/R9 mid--rail divider).

A POWER IN --> +12 V



B U1_SUPPLY

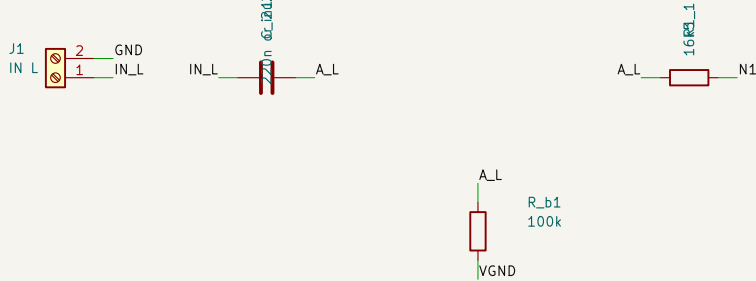


C VIRTUAL GROUND 6 V



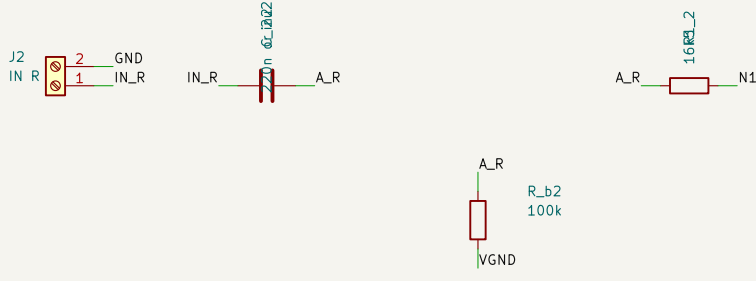
U1 supply: pin 4 = V12, pin 11 = GND. No negative rail.

D INPUT L

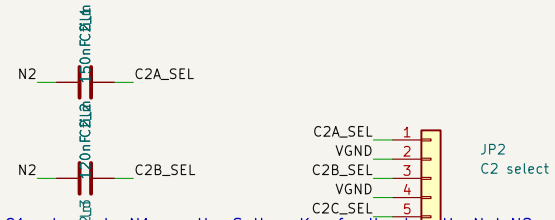
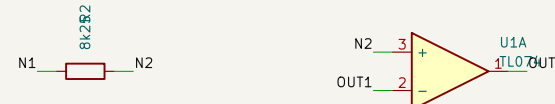
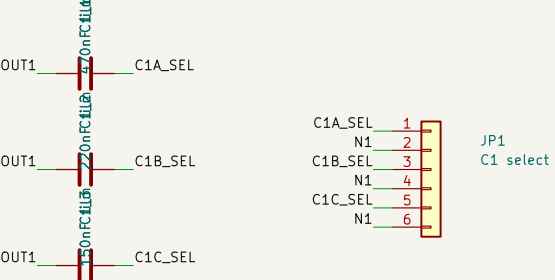


R_b1/R_b2 are the only DC path to A1's + input (C_in blocks DC, C2 is a capacitor). Without them A1 drifts to a rail. They sit after the coupling caps -- 100k at N1 would shift every corner by 8%.

E INPUT R



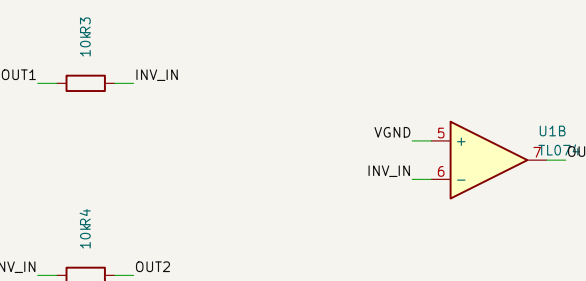
F SALLEN--KEY LOW--PASS



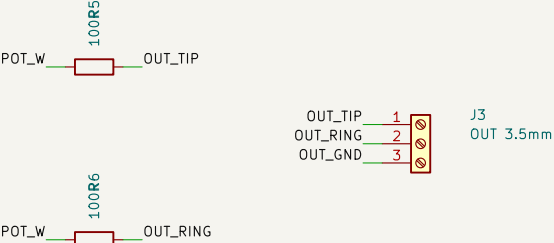
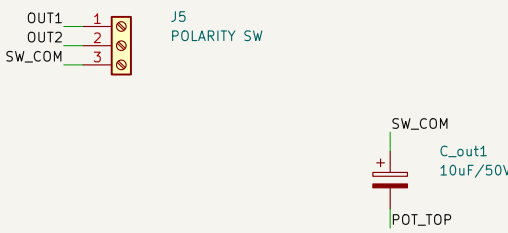
C1 returns to N1 -- the Sallen--Key feedback path. Not N2. JP1 selects C1 (odd pin = cap, even pin = N1). JP2 selects C2 (odd = cap, even = GND). C2 selects INDEPENDENTLY -- nine corners, not three pairs. Corners below are MEASURED on the rev A board, not calculated: the quoted $f_0 = 1/(2\pi R \sqrt{C1 \cdot C2})$ is not a -3 dB point, and C_in sits inside the filter. See Results - Sub Crossover Bring-up.

Intended use is a panel 2--pole 3--position rotary, one pole per header:
pos 1 94 Hz A0-->JP1 even, A1-->JP1.1 (470n) B0-->JP2 even, B1-->JP2.1
pos 2 136 Hz A2-->JP1.5 (150n) B2-->JP2.3
pos 3 189 Hz A3-->JP1.3 (220n) B3-->JP2.5
NEVER wire one header pin to two switch lugs unless both lugs want the same capacitance -- the lugs are then tied together and every position collapses to the same value. This is why C1.1 is fitted rather than paralleling C1.2 || C1.3 on one lug.
Shunts across a pin pair still work as a bench fallback.

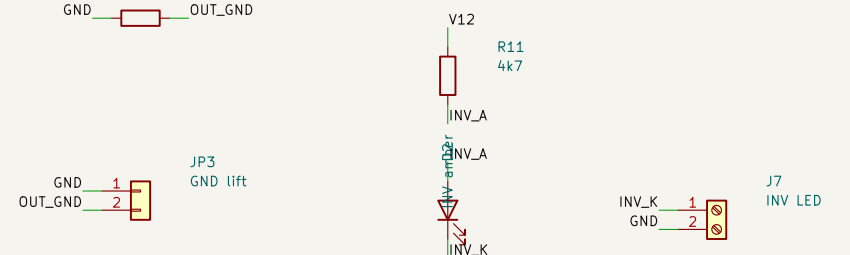
G POLARITY INVERTER



H OUTPUT STAGE



I STATUS LEDS



C_out + faces SW_COM: that node sits at 6 V DC, the pot side at 0 V.
J6 pin 3 = GND, not VGND -- C_out has already removed the 6 V bias; referencing the pot to VGND would thump the driver at power-on.
JP3 shorts R7 (ground lift). Normally fitted.

D1 = power lamp on V12 (proves the LM7812, not just VIN).
D2 = INVERTED lamp, fed from SW2's unused second pole via J7.
Both ~2 mA. No LED touches a signal node.

Off--board: level pot (J6), polarity switch (J5), all panel jacks (J1/J2/J3).
Process: single--sided THT, bottom copper only, board outline 100 x 100 mm.
Copper is isolation--milled on the CNC with an 0.8 mm flat end mill; the fiber laser does the silkscreen only. Routing clearance 0.85 mm / track 1.0 mm.
The 0.8 mm tool width is a hard floor on every copper--to--copper gap. Fixed--pitch pads sit below the 0.85 mm target and carry a kicad.dru exception: DIP--14 0.94 mm (passes), 2.54 mm headers and the TD--220 0.84 mm (do not).
Drills: 0.8 mm x52, 1.0 x26, 1.1 x3, 1.2 x4, 1.5 x15.

Reference names vs the handoff document: KiCad treats a reference with no trailing digit as unannotated, so C1a/C1b/C1c --> C1.1/C1.2/C1.3, C2a/C2b/C2c --> C2.1/C2.2/C2.3, R1a/R1b --> R1.1/R1.2, C_out --> C_out1.

Single--sided THT, CNC isolation milling (0.8 mm end mill), laser silkscreen
Mono--summing 2nd--order Sallen--Key low--pass, single +12 V rail

Sheet: /
File: subxo.kicad_sch

Title: Bose Companion 5 sub crossover

Size: A2 Date: 2026--08--16 Rev: B
KiCad E.D.A. 10.0.4 Id: 1/1